Evolutionary-Conditioned Graph-Based AMP Generation with Molecular Mimicry Detection

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Discovering highly potent antimicrobial peptides with deep generative model HydrAMP

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Antimicrobial Peptides: The Game-Changer in the Epic Battle Against Multidrug-Resistant Bacteria

by Helal F. Hetta [1], Nizar Sirag [2], Shumukh M. Alsharif [1,*], Ahmad A. Alharbi [1], Tala T. Alkindy [1], Alanoud Alkhamali [3], Abdullah S. Albalawi [3], Yasmin N. Ramadan [4], Zainab I. Rashed [4] and Fawaz E. Alanazi [5,*]

AutoGuard

Jakub Giezgała

Hierarchical Conditioning of Diffusion Models Using Tree-of-Life for Studying Species Evolution

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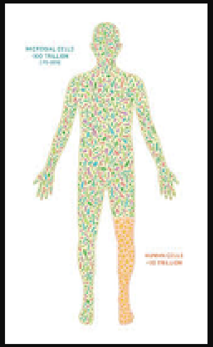
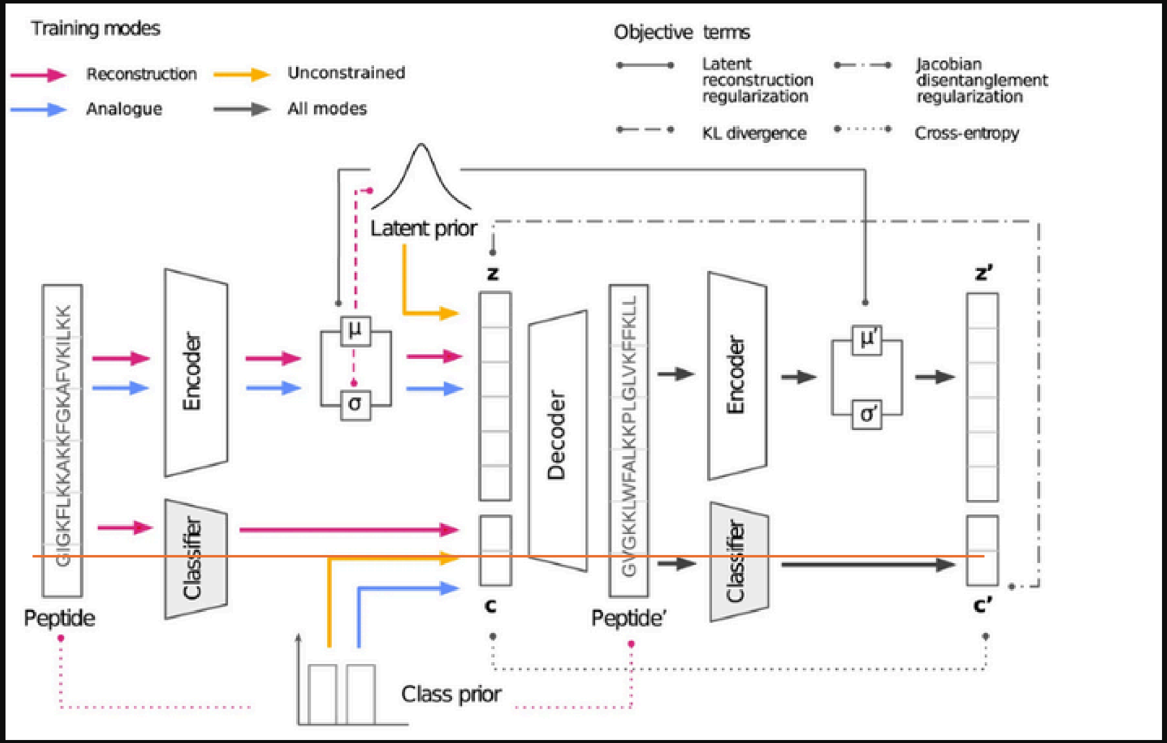
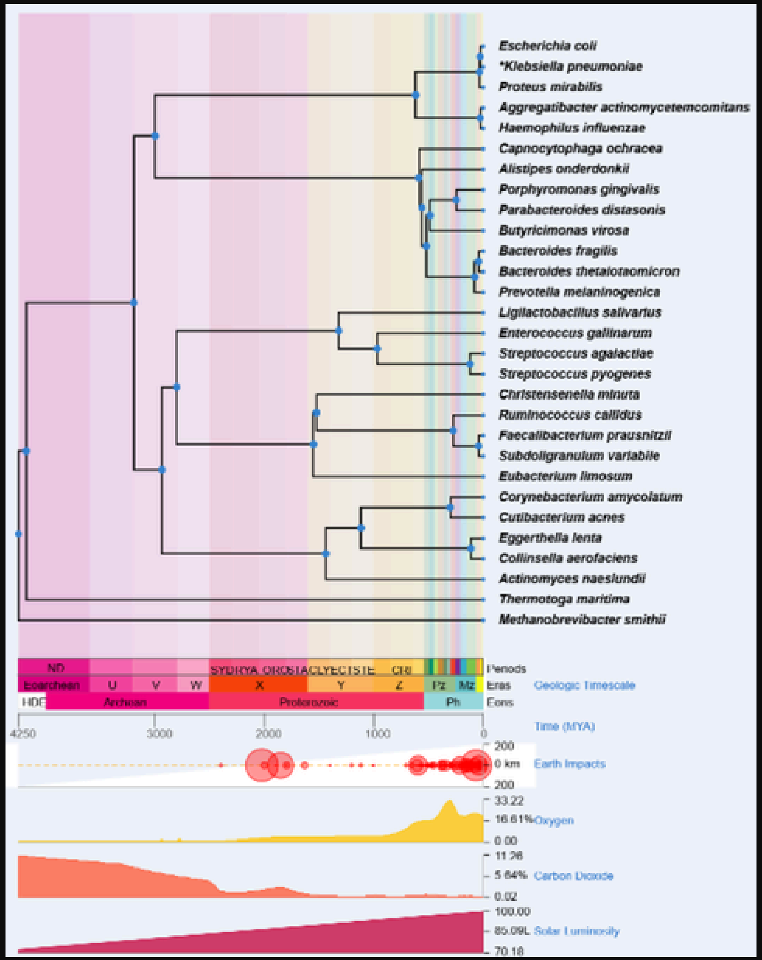
PROBLEMS => MOTIVATION

1. The biological context
2. Data Representation







Introduction and presentation outline

DATA ACQUISITION

EVOLUTIANARY CONDITIONED

AUTOGUARD MODEL

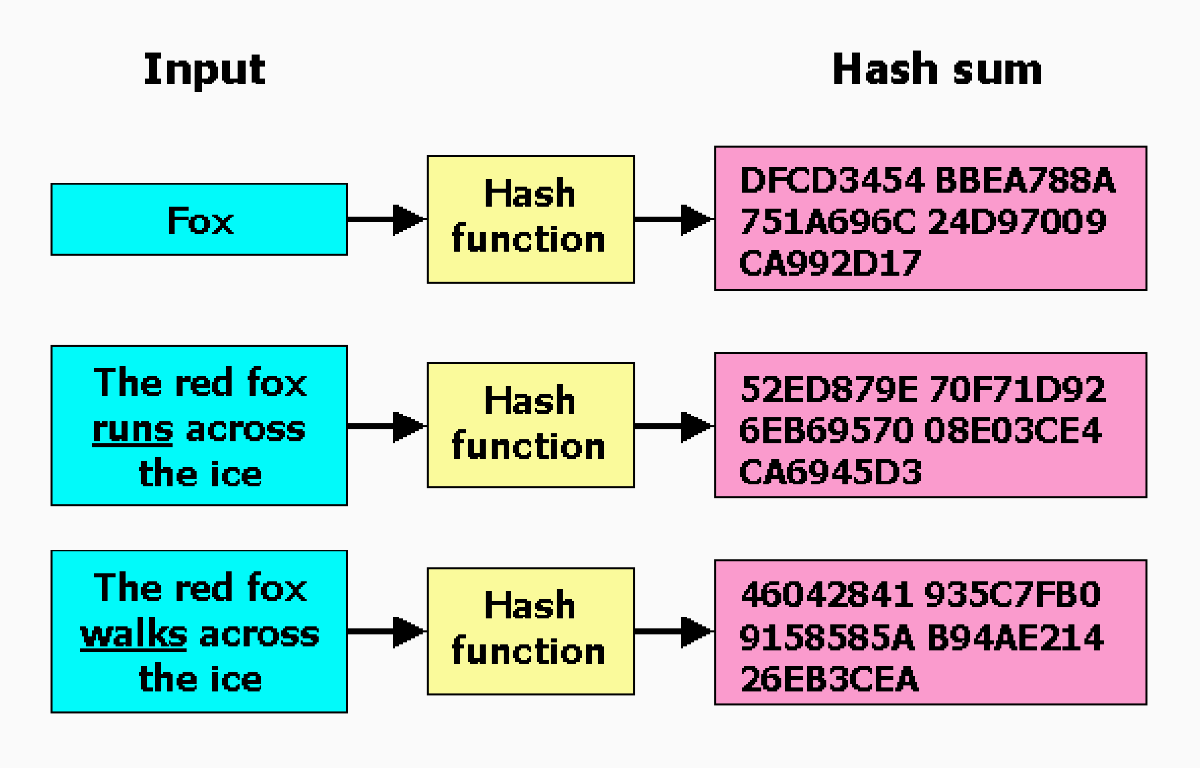
MIMICRY DETECTION

METHODOLOGY & RESULTS

DISCUSSION

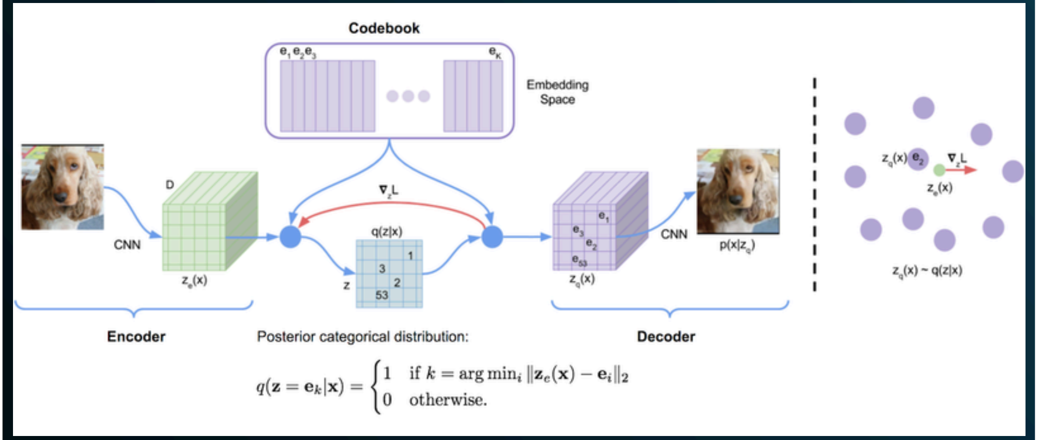
DATA OVERVIEW

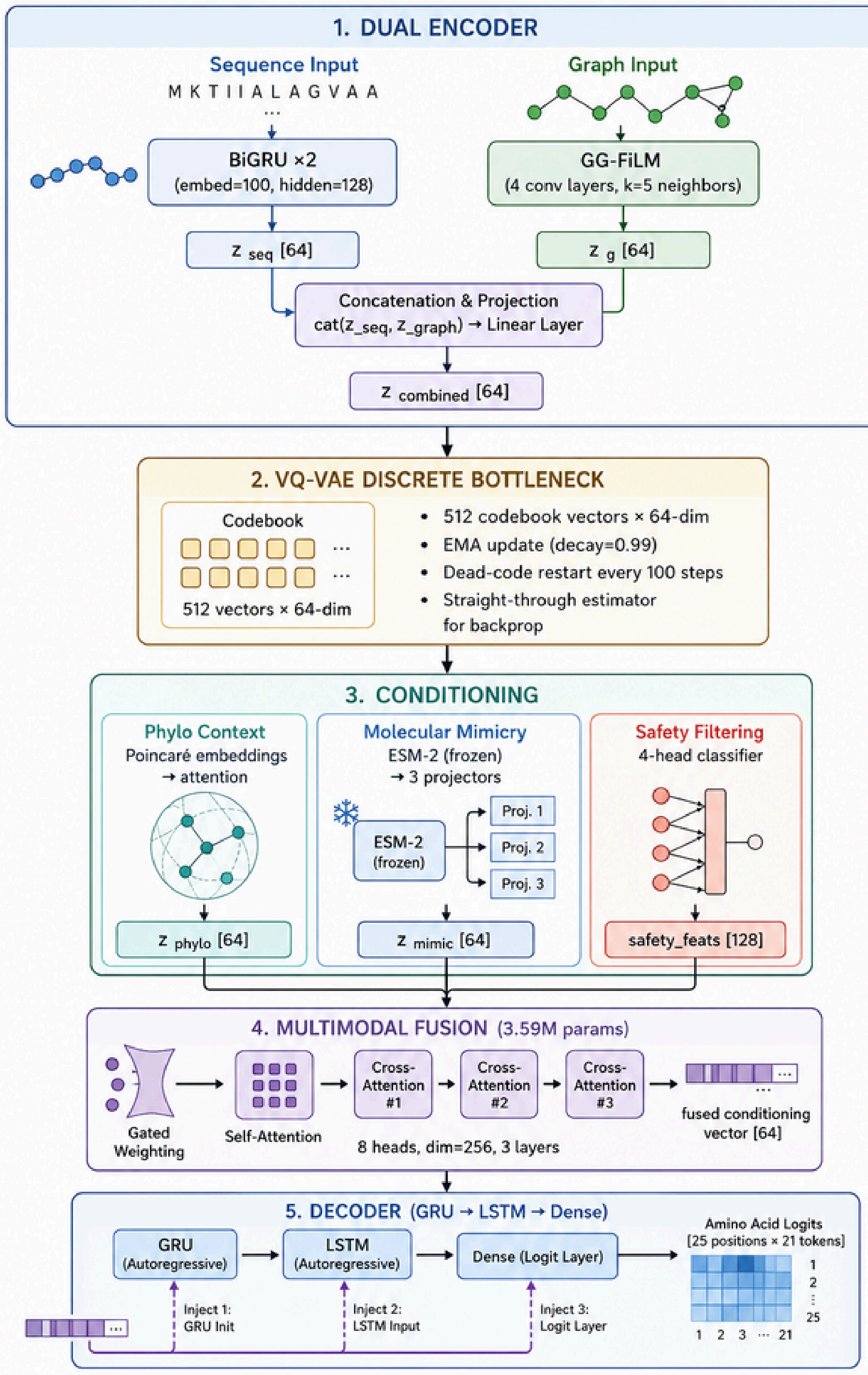
- GRAMPA
- APD6
- DRAMP
- DBAASP
- ToxinPred
- HemoPI
- IEDB
- UniProt



TOTAL SEQUENCES = 64620

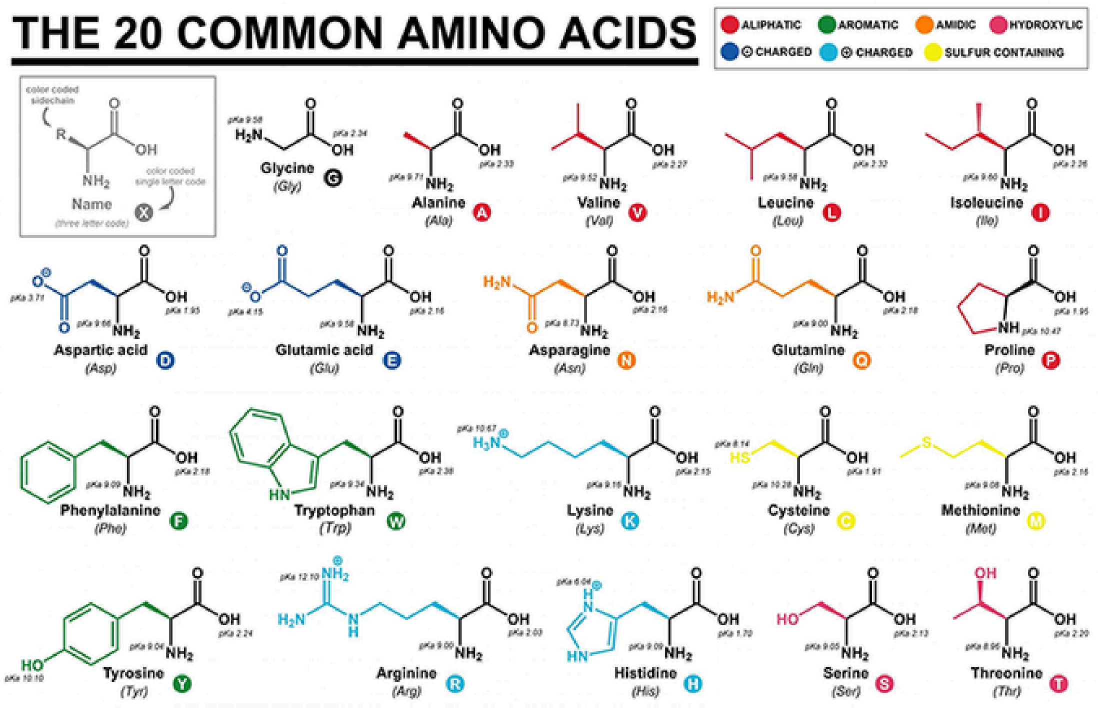
AUTOGUARD MODEL

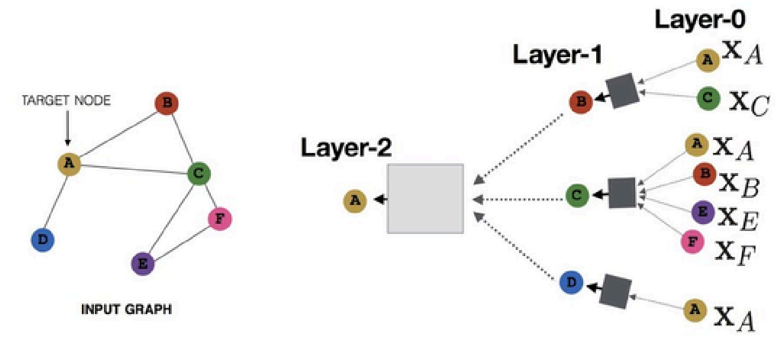




Why Graphs for Peptides?

- 8 PHYSICOCHEMICAL PROPERTIES
- 20-DIM ONE-HOT AMINO ACID IDENTITY
- 8-DIM SINUSOIDAL POSITIONAL ENCODING
- A-HELIX HYDROGEN BOND PARTNERS

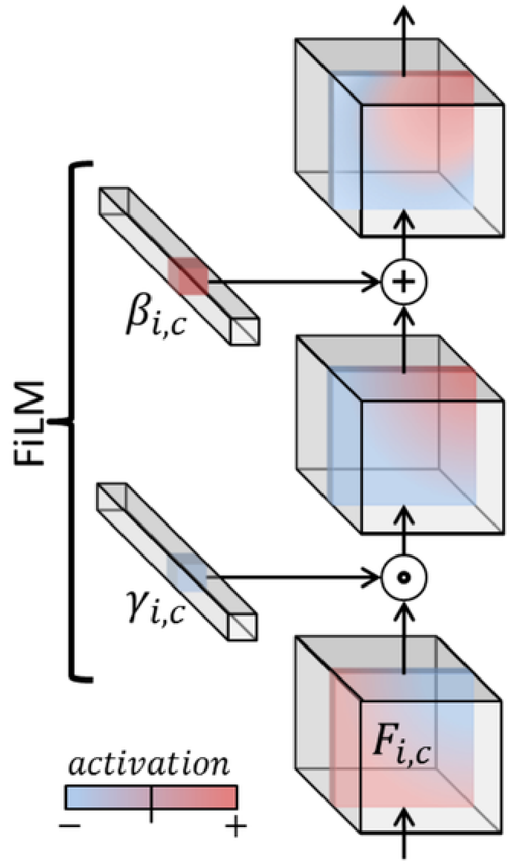




PyTorch geometric

$$\mathcal{L} = \underbrace{1.0 \cdot \mathcal{L}_{recon}}_{CE} + \underbrace{0.25 \cdot \mathcal{L}_{VQ}}_{commitment} + \underbrace{\beta(t) \cdot \mathcal{L}_{KL}}_{annealed} + \underbrace{0.1 \cdot \mathcal{L}_{AMP}}_{classifier} + \underbrace{0.15 \cdot \mathcal{L}_{phylo}}_{conditioning} + \underbrace{0.2 \cdot \mathcal{L}_{mimicry}}_{SLE\ penalty} + \underbrace{0.15 \cdot \mathcal{L}_{safety}}_{tox/hemo}$$

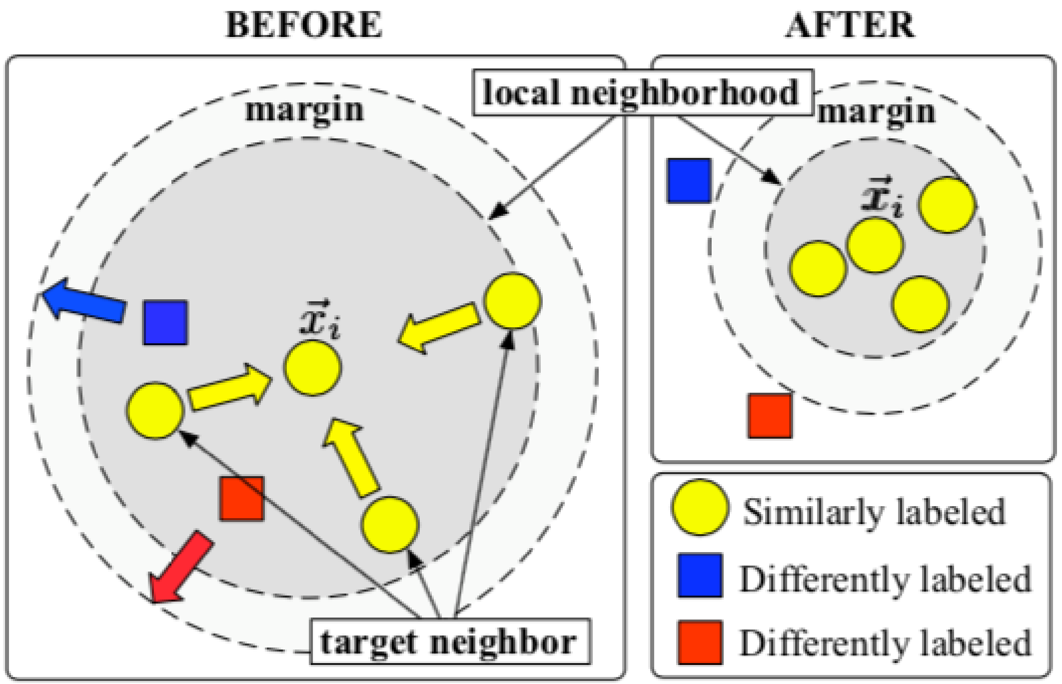
LOSS

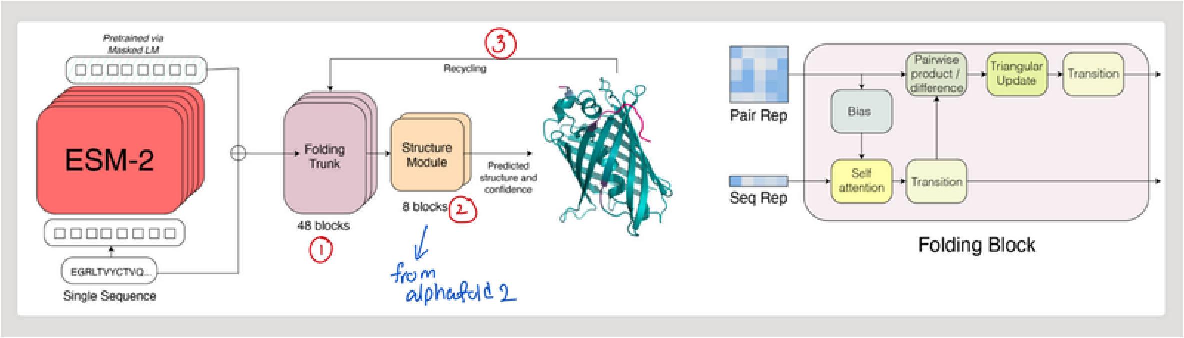


$$h' = \gamma_{phylo} \cdot \mathrm{Conv}(h) + \beta_{phylo}$$

$$\mathcal{L}_{mimicry} = \underbrace{-\mathrm{mean}(\min(s_{pos}, 0.8/\tau))}_{\text{attract to defensins}} + \underbrace{\mathrm{mean}(\mathrm{ReLU}(s_{neg} - 0.5))}_{\text{repel from autoantigens}}$$

SLE Defense





RESULTS

- IN SILICO ONLY
- Phylogenetic conditioning effective only for GRAMPA species (13.4% of training data
- ESM-2 is frozen

Well-developed and original proposal integrating phylogenomics, contrastive mimicry detection, and a VQ-VAE architecture; the motivation for discrete latent space is well-argued (but is it feasible?). Is the project doable in the course timeframe? Too long proposal.

Sequence	Pr(AMP)	Pr(MIC)	Length	Charge	Hydrophobicity	Hydrophobic moment	Plot
ICIFCCGCCHRSKCGMCCKT (27 analogues)	1.00	0.06	20	3.554	0.089	0.009	→
FCIFCCGCCIRGKCGMCCKH	1.00	1.00	20	3.554	0.185	0.072	→
FCIFCFGCCHRSKCGMCCWT	1.00	0.10	20	2.622	0.240	0.070	→
FHIFHCGCVIRGKCGMCCKT	1.00	1.00	20	3.855	0.174	0.088	→
FHTFCFGCFCRGKCGMCCPL	1.00	0.97	20	2.689	0.270	0.160	→
HCIFCFGCCHRSKCGMCVKH	1.00	0.43	20	3.887	0.068	0.138	→
ICHFCFGCCIRTKCAMCCKI	1.00	0.85	20	3.621	0.221	0.129	→
ICICCCGCCHRGKCGMCFKT	1.00	0.98	20	3.554	0.123	0.005	→
ICICCCGVCTRSKCGMCCKT	1.00	0.78	20	3.455	0.102	0.019	→
ICICCFGCFHRSNCGMCCKI	1.00	0.08	20	2.622	0.243	0.224	→
ICIFCCGCFHRSKCGMCCKL	1.00	0.34	20	3.621	0.192	0.093	→
ICIFCFGCCFRTKCGMCCKI	1.00	0.53	20	3.523	0.294	0.157	→
ICIFCFGVCHRSKCGMCFKI	1.00	0.98	20	3.756	0.294	0.040	→
ICIFFCGFCHRSNCGMCCKI	1.00	0.99	20	2.689	0.289	0.116	→

That's all from me. Now, let's discuss it